

YTOCOR 10MG/40MG TABLET

I. DESCRIPTION

YTOCOR 10mg/40mg: White to off-white, 14mm capsule shaped, with '40' marking on one side and '10' on the other side, uncoated tablet.

II. COMPOSITION

YTOCOR 10mg/40mg: This tablet contains 40mg of Simvastatin and 10mg of Ezetimibe.

III. PHARMACODYNAMICS

YTOCOR (ezetimibe/simvastatin) is a lipid-lowering product that selectively inhibits the intestinal absorption of cholesterol and related plant sterols and inhibits the endogenous synthesis of cholesterol.

Mechanism of Action

YTOCOR

Contains ezetimibe and simvastatin, two lipid-lowering compounds with complementary mechanisms of action. YTOCOR reduces elevated total-C, LDL-C, APO B, TG, and non-HDL-C, and increases HDL-C through dual inhibition of cholesterol absorption and synthesis.

Ezetimibe

Ezetimibe inhibits the intestinal absorption of cholesterol. Ezetimibe is orally active and has a mechanism of action that differs from other classes of cholesterol-reducing compounds (e.g. statins, bile acid sequestrants [resins], fibric acid derivatives, and plant stanols). The molecular target of ezetimibe is the sterol transporter, Niemann-Pick C1-Like 1 (NPC1L1), which is responsible for the intestinal uptake of cholesterol and phytosterols.

Ezetimibe localizes at the brush border of the small intestine and inhibits the absorption of cholesterol, leading to a decrease in the delivery of intestinal cholesterol to the liver; statins reduce cholesterol synthesis in the liver and together these distinct mechanisms provide complementary cholesterol reduction.

Simvastatin

After oral ingestion, simvastatin, which is an inactive lactone, is hydrolysed in the liver to the corresponding active β -hydroxyacid form which has a potent activity in inhibiting HMG-CoA reductase (3 hydroxy- 3 methylglutaryl CoA reductase). This enzyme catalyses the conversion of HMG- CoA to mevalonate, an early and rate -limiting step in the biosynthesis of cholesterol.

Simvastatin has been shown to reduce both normal and elevated LDL-C concentrations. LDL is formed from very-low-density protein (VLDL) and is catabolized predominantly by the high affinity LDL receptor. The mechanism of the LDL-lowering effect of simvastatin may involve both reduction of VLDL-cholesterol (VLDL-C) concentration and induction of the LDL receptor, leading to reduced production and increased catabolism of LDL-C. Apolipoprotein B also falls substantially during the treatment with simvastatin. In addition, simvastatin moderately increases HDL-C and reduces plasma TG. As a result of these changes, the ratios of total - to HDL-C and LDL-to HDL-C are reduced.

IV. PHARMACOKINETICS

Absorption

Ezetimibe

After oral administration, ezetimibe is rapidly absorbed and extensively conjugated to a pharmacologically active phenolic glucuronide (ezetimibe-glucuronide). Mean maximum plasma concentrations (C_{max}) occur within 1 to 2 hours for the ezetimibe-glucuronide and 4 to 12 hours for ezetimibe. The absolute bioavailability of ezetimibe cannot be determined as the compound is virtually insoluble in aqueous media suitable for injection.

Concomitant food administration (high fat or non-fat meals) had no effect on the oral bioavailability of ezetimibe when administered as ezetimibe 10mg tablets.

Simvastatin

The availability of β -hydroxyacid to the systemic circulation following an oral dose of simvastatin was found to be less than 5% of the dose, consistent with extensive hepatic first-pass extraction. The major metabolites of simvastatin present in

human plasma are the β -hydroxyacid and four additional active metabolites.

Relative to the fasting state, the plasma profiles of both active and total inhibitors were not affected when simvastatin was administered immediately before a test meal.

Distribution

Ezetimibe

Ezetimibe and ezetimibe-glucuronide are bound 99.7% and 88 to 92% to human plasma proteins, respectively.

Simvastatin

Both simvastatin and the β -hydroxyacid are bound to human plasma proteins (95%).

The pharmacokinetics of single and multiple doses of simvastatin showed that no accumulation of drug occurred after multiple dosing. The maximum plasma concentration of inhibitors occurred 1.3 to 2.4 hours post-dose.

Metabolism

Ezetimibe

Ezetimibe is metabolized primarily in the small intestine and liver via glucuronide conjugation (a phase (II) reaction) with subsequent biliary excretion. Minimal oxidative metabolism (a phase (I) reaction) has been observed in all species evaluated. Ezetimibe and ezetimibe-glucuronide are the major drug-derived compounds detected in plasma, constituting approximately 10 to 20% and 80 to 90 % of the total drug in plasma, respectively. Both ezetimibe and ezetimibe-glucuronide are slowly eliminated for plasma with evidence of significant enterohepatic recycling. The half-life for ezetimibe and ezetimibe-glucuronide is approximately 22 hours.

Simvastatin

Simvastatin is an inactive lactone which is readily hydrolysed *in vivo* to the corresponding β -hydroxyacid, a potent inhibitor of HMG-CoA reductase. Hydrolysis takes place mainly in liver, the rate of hydrolysis in human plasma is very low.

In man simvastatin is well absorbed and undergoes extensive hepatic first-pass extraction. The extraction in the liver is dependent on the hepatic blood flow. The liver is its primary site of action, with subsequent excretion of drug equivalents in the bile. Consequently, availability of active drug to the systemic circulation is low.

Following an intravenous injection of the β -hydroxyacid metabolite, its half-life averaged 1.9 hours.

Elimination

Ezetimibe

Following oral administration of ^{14}C -ezetimibe (20 mg) to human subjects, total ezetimibe accounted for approximately 93 % of the total radioactivity in plasma. Approximately 78% and 11% of the administered radioactivity were recovered in the faeces and urine, respectively, over a 10-day collection period. After 48 hours, there were no detectable levels of radioactivity in the plasma.

Simvastatin

Following an oral dose of radioactive simvastatin to man, 13% of the radioactivity was excreted in the urine and 60% in the faeces within 96 hours. The amount recovered in the faeces represents absorbed drug equivalents excreted in bile as well as unabsorbed drug. Following an intravenous injection of the β -hydroxyacid metabolite, an average of only 0.3% of the IV dose was excreted in urine as inhibitors.

V. INDICATION

Prevention of Cardiovascular Events

VYTORCOR is indicated to reduce the risk of cardiovascular events (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, hospitalization for unstable angina, or need for revascularization), in patients with coronary heart disease (CHD) and history of acute coronary syndrome (ACS), either previously treated with a statin or not.

Primary Hypercholesterolemia

VYTORCOR is indicated as adjunctive therapy to diet for the reduction of elevated total cholesterol (total-C), low density lipoprotein cholesterol (LDL-C), apolipoprotein B (Apo B), triglycerides (TG), and non-high density lipoprotein cholesterol (non-HDL-C), and to increase high-density lipoprotein cholesterol (HDL-C) in patients with primary (heterozygous familial and non-familial) hypercholesterolemia or mixed hyperlipidaemia.

Homozygous Familial Hypercholesterolemia (HoFH)

VYTOCOR is indicated for the reduction of elevated total-C and LDL-C levels in patients with HoFH. Patients may also receive adjunctive treatments (e.g. LDL apheresis).

VI. DOSAGE AND ADMINISTRATION

The patient should be placed on a standard cholesterol-lowering diet before receiving VYTOCOR and should continue on this diet during treatment with VYTOCOR. The dosage should be individualized according to the baseline LDL-C level, the recommended goal of therapy, and the patient's response. VYTOCOR should be taken as a single daily dose in the evening, with or without food.

The dosage range is 10/10mg/day through 10/80mg/day. The recommended usual starting dose is 10/20mg/day. Initiation of therapy with 10/10mg/day may be considered for patients requiring less aggressive LDL-C reductions. Patients who require a larger reduction in LDL-C (greater than 55%) may be started at 10/40mg/day. After initiation or titration of VYTOCOR, lipid levels may be analysed after 2 or more weeks and dosage adjusted, if needed.

The 10/80mg dose of ezetimibe/simvastatin is only recommended in patients at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks. (*see IX. WARNING(s) AND PRECAUTION(s)*).

For patients who are currently on both simvastatin and ezetimibe, no dosage adjustment is required when switching to VYTOCOR.

Patients with Primary Hypercholesterolemia

In patients with primary hyperlipidemia or mixed hyperlipidemia, the dosage range is 10/10 mg/day through 10/80 mg/day. The recommended usual starting dose is 10/20 mg/day. Initiation of therapy with 10/10 mg/day may be considered for patients requiring less aggressive LDL-C reductions. Patients who require a larger reduction in LDL-C (greater than 55%) may be started at 10/40 mg/day. After initiation or titration of VYTOCOR, lipid levels may be analyzed after 2 or more weeks and dosage adjusted, if needed. The 10/80 mg dose of VYTOCOR is only recommended in patients at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks (*see IX. WARNING(s) AND PRECAUTION(s)*).

Patients with Coronary Heart Disease

The starting dose may be initiated with 10/40 mg once a day in the evening. The 10/80-mg dose is only recommended when the benefits are expected to outweigh the potential risks. (*see IX. WARNING(s) AND PRECAUTION(s)*).

Dosage in Patients with Homozygous Familial Hypercholesterolemia

The recommended dosage for patients with homozygous familial hypercholesterolemia is VYTOCOR 10/40mg/day or 10/80mg/day in the evening. The 10/80mg dose is only recommended when the benefits are expected to outweigh the potential risks (*see above, VIII. CONTRAINDICATIONS, and IX. WARNING(s) AND PRECAUTION(s)*). VYTOCOR should be used as an adjunct to other lipid-lowering treatments (e.g. LDL apheresis) in these patients or if such treatments are unavailable. In patients taking lomitapide concomitantly with VYTOCOR, the dose of VYTOCOR should not exceed 10/40mg/day (*see IX. WARNING(s) AND PRECAUTION(s) and X. DRUG INTERACTIONS*).

Use in Special Populations

Geriatric Patients

No dosage adjustment is required for elderly patients.

Plasma concentrations for total ezetimibe are about 2-fold higher in the elderly (≥65 years) than in the young (18 to 45 years). LDL-C reduction and safety profile are comparable between elderly and young subjects treated with ezetimibe.

Paediatric Patients

Treatment with VYTOCOR is **NOT recommended**.

The absorption and metabolism of ezetimibe are similar between children and adolescents (10 to 18 years) and adults. Based on total ezetimibe, there are no pharmacokinetic differences between adolescents and adults. Pharmacokinetic data in paediatric population < 10 years of age are not available.

Use in Hepatic Impairment

No dosage adjustment is required in patients with mild hepatic insufficiency (Child-Pugh score 5 or 6).

Treatment with VYTOCOR is not recommended in patients with moderate (Child-Pugh score 7 to 9) or severe (Child-Pugh score > 9) liver dysfunction (*see IX. WARNING(s) AND PRECAUTION(s)*).

Use in Renal Impairment

No dosage adjustment is required for patients with moderate renal insufficiency. If treatment in patients with severe renal insufficiency (creatinine clearance < 30ml/min) is deemed necessary, dosages above 10/10mg/day should be implemented cautiously.

Co-administration with other medicines

Dosing of VYTOCOR should occur 2 hours before or 4 hours after administration of a bile acid sequestrant.

In patients taking fibrates (other than gemfibrozil and fenofibrate) concomitantly with VYTOCOR, the dose of VYTOCOR should not exceed 10/10mg/day. (*see IX. WARNING(s) AND PRECAUTION(s) and X. DRUG INTERACTIONS*).

In patients taking amiodarone, verapamil or diltiazem concomitantly with VYTOCOR, the dose of VYTOCOR should not exceed 10/20mg/day (*see IX. WARNING(s) AND PRECAUTION(s) and X. DRUG INTERACTIONS*).

In patients taking amlodipine or lipid-lowering dose of niacin ($\geq 1\text{g/day}$) concomitantly with VYTOCOR, the dose of VYTOCOR should not exceed 10/40mg/day (*see IX. WARNING(s) AND PRECAUTION(s) and X. DRUG INTERACTIONS*).

VII. ROUTE OF ADMINISTRATION

Oral

VIII. CONTRAINDICATIONS

- Hypersensitivity to the active substances or to any of the excipients.
- Active liver disease or unexplained persistent elevations of serum transaminases.
- Pregnancy and nursing (*see XI. PREGNANCY AND LACTATION*)
- Concomitant administration of potent CYP3A4 inhibitors (e.g. itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors, boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin, nefazodone, and drugs containing cobicistat) (*see IX. WARNING(s) AND PRECAUTION(s) and X. DRUG INTERACTIONS*).
- Concomitant administration of gemfibrozil, cyclosporine, or danazol (*see IX. WARNING(s) AND PRECAUTION(s) and X. DRUG INTERACTIONS*).

IX. WARNING(s) AND PRECAUTION(s)

Myopathy/Rhabdomyolysis

Simvastatin, like other inhibitors of HMG-CoA reductase, occasionally causes myopathy manifested as muscle pain, tenderness, or weakness with creatine kinase (CK) above 10X the upper limit of normal (ULN). Myopathy sometimes takes the form of rhabdomyolysis with or without acute renal failure secondary to myoglobinuria, and rare fatalities have occurred. The risk of myopathy is increased by high levels of HMG-CoA reductase inhibitory activity in plasma. Predisposing factors for myopathy include advanced age (≥ 65 years), female gender, uncontrolled hypothyroidism, and renal impairment.

The risk of myopathy is greater in patients on simvastatin 80mg compared with other statin-based therapies with similar LDL-C lowering efficacy. Therefore, 10/80mg dose of VYTORCOR should only be used in patients at high risk for cardiovascular complications who have not achieved their treatment goals on lower doses and when the benefits are expected to outweigh the potential risks. In patients taking VYTORCOR 10/80mg for whom an interacting agent is needed, a lower dose of VYTORCOR or an alternative statin-ezetimibe regimen with less potential for drug-drug interactions should be used (*see below, VI.DOSAGE AND ADMINISTRATION and VIII.CONTRAINDICATIONS*).

All patients starting therapy with VYTORCOR, or whose dose of VYTORCOR is being increased, should be advised of the risk of myopathy and told to report promptly and unexplained muscle pain, tenderness or weakness. VYTORCOR therapy should be discontinued immediately if myopathy is diagnosed or suspected.

The presence of these symptoms, and a CK level >10 times the upper limit of normal indicates myopathy. In most cases, when patients were promptly discontinued from simvastatin treatment, muscle symptoms and CK increases resolved (*see XII.SIDE EFFECTS*). Periodic CK determinations may be considered in patients starting therapy with VYTORCOR or whose dose is being increased. Periodic CK determinations are recommended for patients titrating to the 10/80mg dose. There is no assurance that such monitoring will prevent myopathy.

Many of the patients who have developed rhabdomyolysis on therapy with simvastatin have had complicated medical histories, including renal insufficiency usually as a consequence of long-standing diabetes mellitus. Such patients taking VYTORCOR merit closer monitoring. Therapy with VYTORCOR should be temporarily stopped a few days prior to elective major surgery and when any major medical or surgical condition supervenes.

Use in the elderly

Because advanced age (≥65 years) is a predisposing factor for myopathy, VYTORCOR should be prescribed with caution in the elderly. In a clinical trial of patients treated with simvastatin 80 mg/day, patients ≥65 years of age had an increased risk of myopathy compared to patients <65 years of age.

Because VYTORCOR contains simvastatin, the risk of myopathy/rhabdomyolysis is increase by concomitant use of VYTORCOR with following drugs:

Contraindicated Drugs

- **Potent inhibitors of CYP3A4:** Concomitant use with medicines labelled as having potent inhibitory effect on CYP3A4 at therapeutic doses (e.g. **itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors, boceprevir, telaprevir, nefazodone, or drugs containing cobicistat**) is contraindicated. If short-term treatment with potent CYP3A4 inhibitors is unavoidable, therapy with VYTORCOR should be suspended during course of treatment. (*See VIII.CONTRAINDICATIONS, X.DRUG INTERACTIONS*).
- **Gemfibrozil, cyclosporine, or danazol:** Concomitant use of these drugs with VYTORCOR is contraindicated. (*See VIII.CONTRAINDICATIONS and X.DRUG INTERACTIONS*).

Other Drugs

- **Fusidic acid:** Patients on fusidic acid treated concomitantly with simvastatin may have an increased risk of myopathy/rhabdomyolysis. Co-administration with fusidic acid is not recommended. In patients where the use of systemic fusidic acid is considered essential, VYTORCOR should be discontinued throughout the duration of fusidic acid treatment. In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g. for the treatment of severe infections, the need for co-administration of VYTORCOR and fusidic acid should only be considered on a case-by-case basis under close medical supervision.
- **Amiodarone:**
The risk of myopathy including rhabdomyolysis is increased by concomitant administration of amiodarone with VYTORCOR. The dose of VYTORCOR should not exceed 10/20mg daily in patients receiving concomitant medication with amiodarone (*see VI.DOSAGE AND ADMINISTRATION and X.DRUG INTERACTIONS*).

- **Calcium Channel Blockers**

- **Verapamil or diltiazem:** Patients on diltiazem treated concomitantly with simvastatin 80mg had an increased risk of myopathy. The dose of VYTOCOR should not exceed 10/20mg daily in patients receiving concomitant medication with verapamil or diltiazem (*See X.DRUG INTERACTIONS*).
 - **Amlodipine:** Patients on amlodipine treated concomitantly with simvastatin 80mg had a slightly increased risk of myopathy. The dose of VYTOCOR should not exceed 10/40mg daily in patients receiving concomitant medication with amlodipine (*See X.DRUG INTERACTIONS*).
- **Lomitapide:** The dose of VYTOCOR should not exceed 10/40mg daily in patients with HoFH receiving concomitant medication with lomitapide (*See X.DRUG INTERACTIONS*).

- **Moderate inhibitors of CYP3A4:** Patients taking other medicines labelled as having a moderate inhibitory effect of CYP3A4 concomitantly with VYTOCOR, particularly higher VYTOCOR doses, may have an increased risk of myopathy. When coadministering VYTOCOR with a moderate inhibitor of CYP3A4, a dose adjustment of VYTOCOR may be necessary.
- **Other Fibrates:** The safety and effectiveness of VYTOCOR administered with fibrates, have not been studied. Therefore, the concomitant use of VYTOCOR and fibrates, should be avoided. Concomitant use of gemfibrozil is contraindicated (*see VIII.CONTRAINDICATIONS*).
- **Niacin (≥1g/day) :** The benefit of combined use of simvastatin with niacin should be carefully weighed against the potential risks of the combination. The dose of simvastatin should not exceed 40mg daily in patients receiving concomitant medication with niacin (nicotinic acid) ≥1g/day. Cases of myopathy/rhabdomyolysis have been observed with simvastatin co-administered with lipid-modifying doses (≥1g/day) of niacin (*See X.DRUG INTERACTIONS*).
- **Anticoagulants:** If VYTOCOR is added to warfarin, another coumarin anticoagulant, or flutidione, the International Normalized Ratio (INR) should be appropriately monitored (*See X.DRUG INTERACTIONS*).

Liver Enzymes

It is recommended that LFTs be performed before treatment with VYTOCOR begins and thereafter when clinically indicated. Patients titrated to 10/80mg dose should receive an additional test prior to titration, 3 months after titration to the 10/80mg dose, and periodically thereafter (e.g. semi-annually) for the first year of treatment. Special attention should be paid to patients who develop elevated serum transaminase levels, and in these patients, measurements should be repeated promptly and then performed more frequently. If the transaminase levels show evidence of progression, particularly if they rise 3X ULN and are persistent, the drug should be discontinued. Note that ALT may emanate from muscle, therefore ALT rising with CK may indicate myopathy (*see IX.WARNING(s) AND PRECAUTION(s)*).

There have been rare reports of fatal and non-fatal hepatic failure in patients taking statins, including simvastatin. If serious liver injury with clinical symptoms and/or hyperbilirubinemia or jaundice occurs during treatment with VYTOCOR, promptly interrupt therapy. If an alternate aetiology is not found, do not restart VYTOCOR.

VYTOCOR should be used with caution in patients who consume substantial quantities of alcohol and/or have a past history of liver disease. Active liver disease or unexplained persistent transaminase elevations are contraindications to the use of VYTOCOR.

Hepatic Insufficiency

Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe hepatic insufficiency, VYTOCOR is not recommended in these patients (*see VI.DOSAGE AND ADMINISTRATION (use in special populations)*).

X. **DRUG INTERACTIONS**

No clinically significant pharmacokinetics interaction was seen when ezetimibe was co-administered with simvastatin. VYTOR is bioequivalent to co-administered ezetimibe and simvastatin.

Contraindicated Drugs

Concomitant use of the following drugs is **contraindicated** :-

• **Potent Inhibitors of CYP3A4**

No clinically significant pharmacokinetic interactions have been observed between ezetimibe and drugs known to be metabolised by cytochromes P450 1A2, 2D6, 2C8, 2C9, and 3A4, or N-acetyltransferase. Simvastatin is metabolised by CYP3A4 but has no CYP3A4 inhibitory activity. Therefore, simvastatin is not expected to affect plasma concentrations of other drugs metabolised by CYP3A4.

Potent inhibitors of CYP3A4 increase the risk of myopathy by reducing the elimination of the simvastatin component of VYTOR. Concomitant use of drugs labelled as having a strong inhibitory effect on CYP3A4 (itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, nefazodone, and drugs containing cobicistat) is contraindicated (*see VIII.CONTRAINDICATIONS and IX.WARNING(s) AND PRECAUTION(s)*). If treatment with potent CYP3A4 inhibitors is unavoidable, therapy with simvastatin should be suspended during the course of treatment.

• **Gemfibrozil, Ciclosporin, or Danazol**

Concomitant use of these drugs with simvastatin is contraindicated (*see VIII.CONTRAINDICATIONS and IX.WARNING(s) AND PRECAUTION(s)*).

Other drug interactions

• **Other fibrates**

The dose of simvastatin should not exceed 10 mg daily in patients receiving concomitant medication with fibrates other than gemfibrozil or fenofibrate. When simvastatin and fenofibrate are given concomitantly, there is no evidence that the risk of myopathy exceeds the sum of the individual risks of each agent. Caution should be used when prescribing fenofibrate with simvastatin, as either agent can cause myopathy when given alone. Addition of fibrates to simvastatin typically provides little additional reduction in LDL-C, but further reductions of TG and further increases in HDL-C may be obtained. Combinations of fibrates with simvastatin have been used without myopathy in small short-term clinical studies with careful monitoring (*see VIII.CONTRAINDICATIONS and IX.WARNING(s) AND PRECAUTION(s)*).

• **Calcium Channel Blockers**

- **Verapamil or diltiazem:** Patients on diltiazem treated concomitantly with simvastatin 80mg had an increased risk of myopathy. The dose of VYTOR should not exceed 10/20mg daily in patients receiving concomitant medication with verapamil or diltiazem (*See IX.WARNING(s) AND PRECAUTION(s)*).
- **Amlodipine:** Patients on amlodipine treated concomitantly with simvastatin 80mg had a slightly increased risk of myopathy. The dose of VYTOR should not exceed 10/40mg daily in patients receiving concomitant medication with amlodipine (*See IX.WARNING(s) AND PRECAUTION(s)*).

• **Fusidic Acid**

The risk of myopathy including rhabdomyolysis may be increased by concomitant administration of fusidic acid with VYTOR (*see VIII.CONTRAINDICATIONS and IX.WARNING(s) AND PRECAUTION(s)*).

• **Amiodarone**

In a clinical trial, myopathy was reported in 6% of patients receiving simvastatin 80mg and amiodarone. The dose of VYTOR should not exceed 10/20mg daily in patients receiving concomitant medication with amiodarone (*see VI.DOSAGE AND ADMINISTRATION and IX.WARNING(s) AND PRECAUTION(s)*).

350 • **Cholestyramine**

351 Concomitant cholestyramine administration decreased the mean area under the curve (AUC) of total ezetimibe (ezetimibe +
352 ezetimibe glucuronide) approximately 55%. The incremental LDL-C reduction due to adding INEGY to cholestyramine may be
353 lessened by this interaction.

354 • **Calcium Channel Blockers**

355 The risk of myopathy including rhabdomyolysis is increased by concomitant administration of verapamil, diltiazem or
356 amlodipine with VYTORCOR (see VI.DOSAGE AND ADMINISTRATION and IX.WARNING(s) AND PRECAUTION(s)).

357 • **Lomitapide**

358 The risk of myopathy and rhabdomyolysis may be increased by concomitant administration of lomitapide with simvastatin
359 (see VI.DOSAGE AND ADMINISTRATION and IX.WARNING(s) AND PRECAUTION(s)). Therefore, in patients with HoFH, the dose
360 of INEGY must not exceed 10/40 mg daily in patients receiving concomitant medication with lomitapide.

361 • **Moderate Inhibitors of CYP3A4**

362 Patients taking other medicines labelled as having a moderate inhibitory effect on CYP3A4 concomitantly with VYTORCOR,
363 particularly higher VYTORCOR doses, may have an increased risk of myopathy (see IX.WARNING(s) AND PRECAUTION(s)).

364 • **Inhibitors of the Transport Protein OATP1B1**

365 Simvastatin acid is a substrate of the transport protein OATP1B1. Concomitant administration of medicinal products that are
366 inhibitors of the transport protein OATP1B1 may lead to increased plasma concentrations of simvastatin acid and an increased
367 risk of myopathy (see VIII.CONTRAINDICATIONS and IX.WARNING(s) AND PRECAUTION(s)).

368 • **Inhibitors of Breast Cancer Resistant Protein (BCRP)**

369 Simvastatin is a substrate of the Breast Cancer Resistant Protein (BCRP) efflux transporter. Concomitant administration of
370 products that are inhibitors of BCRP (e.g., elbasvir and grazoprevir) may lead to increased plasma concentrations of simvastatin
371 and an increased risk of myopathy. When coadministering simvastatin with an inhibitor of BCRP, a dose adjustment of VYTORCOR
372 may be necessary.

373 • **Niacin**

374 The dose of simvastatin should not exceed 40mg daily in patients receiving concomitant medication with niacin (nicotinic
375 acid) $\geq 1\text{g/day}$. Cases of myopathy/rhabdomyolysis have been observed with simvastatin co-administered with lipid-modifying
376 doses ($\geq 1\text{g/day}$) of niacin (see VI.DOSAGE AND ADMINISTRATION and IX.WARNING(s) AND PRECAUTION(s)).

377 • **Colchicine**

378 There have been reports of myopathy and rhabdomyolysis with the concomitant administration of colchicine and simvastatin,
379 in patients with renal impairment. Close clinical monitoring of such patients taking this combination is advised.

380 **Other drug interactions**

381 • **Grapefruit Juice**

382 Grapefruit juice contains one or more components that inhibit CYP3A4 and can increase the plasma concentrations of drugs
383 metabolized by CYP3A4. Intake of grapefruit juice during treatment with VYTORCOR should therefore be avoided.

384 • **Anticoagulants**

385 In patients taking coumarin anticoagulants, prothrombin time should be determined before starting VYTORCOR and frequently
386 enough during early therapy to ensure that no significant alteration of prothrombin time occurs. Once a stable prothrombin
387 time has been documented, prothrombin times can be monitored at the intervals usually recommended for patients on
388 coumarin anticoagulants. If the dose of VYTORCOR is changed or discontinued, the same procedure should be repeated.
389 Simvastatin therapy has not been associated with bleeding or with changes in prothrombin time in patients not taking
390 anticoagulants.

391 The effect of VYTORCOR on the prothrombin time has not been studied.

• **Antacids**

Concomitant antacid administration decreased the rate of absorption of ezetimibe but had no effect on the bioavailability of ezetimibe. This decreased rate of absorption is not considered clinically significant.

XI. PREGNANCY AND LACTATION

Atherosclerosis is a chronic process, and ordinarily discontinuation of lipid-lowering drugs during pregnancy should have little impact on the long-term risk associated with primary hypercholesterolemia.

Use in Pregnancy: VYTOCOR is contraindicated during pregnancy.

Use in Lactation: It is not known whether the active components of VYTOCOR are excreted into human breast milk; therefore, women who are nursing should not take VYTOCOR.

XII. SIDE EFFECTS

VYTOCOR is generally well tolerated.

The following common ($\geq 1/100$, $< 1/10$) or uncommon ($\leq 1/1000$, $< 1/100$); drug-related adverse experiences were reported in patients taking ezetimibe/simvastatin and at a greater incidence than placebo.

Investigations:

Common: ALT and/or AST increased; blood CK increased

Uncommon: blood bilirubin increased; blood uric acid increased; gamma-glutamyltransferase increased; international normalized ratio increased; protein urine present; weight decreased

Nervous system disorders:

Uncommon: dizziness; headache

Gastrointestinal disorders

Uncommon: abdominal pain; abdominal discomfort; abdominal pain upper; dyspepsia; flatulence; nausea; vomiting

Skin and subcutaneous tissue disorders

Uncommon: pruritus; rash

Musculoskeletal and connective tissue disorders

Uncommon: arthralgia; muscle spasms; muscular weakness; musculoskeletal discomfort; neck pain; pain in extremity

General disorders and administration site conditions

Uncommon: asthenia; fatigue; malaise; oedema peripheral

Psychiatric disorders

Uncommon: sleep disorders

The following common ($\geq 1/100$, $< 1/10$) or uncommon ($\leq 1/1000$, $< 1/100$); drug-related adverse experiences were reported in patients taking ezetimibe/simvastatin and at a greater incidence than statins administered alone.

Investigations:

Common: ALT and/or AST increased

Uncommon: blood bilirubin increased; blood CK increased; gamma-glutamyltransferase increased

Nervous system disorders:

Uncommon: headache; paraesthesia

Gastrointestinal disorders

Uncommon: abdominal distension; diarrhoea; dry mouth; dyspepsia; flatulence; gastroesophageal reflux disease; vomiting

Skin and subcutaneous tissue disorders

Uncommon: pruritus; rash; urticaria

Musculoskeletal and connective tissue disorders

Common: myalgia

Uncommon: arthralgia; back pain; muscle spasms; muscular weakness; musculoskeletal pain; pain in extremity

General disorders and administration site conditions

Uncommon: asthenia; chest pain; fatigue; oedema peripheral

Psychiatric disorders

Uncommon: insomnia

XIII. SYMPTOMS AND TREATMENT OF OVERDOSAGE

No specific treatment of overdosage with VYTOCOR can be recommended. In the event of overdose, symptomatic and supportive measures should be employed.

Ezetimibe

A few cases of overdosage have been reported; most have not been associated with adverse experiences. Reported adverse experiences have not been serious.

Simvastatin

A few cases of overdosage have been reported; the maximum dose taken was 3.6 g. All patients recovered without sequelae.

XIV. EFFECTS ON ABILITY TO DRIVE & USE MACHINE

No studies on the effects on the ability to drive and use machines have been performed. However, when driving vehicles or operating machines, it should be taken into account that dizziness has been reported.

XV. STORAGE CONDITION

Store below 30°C.

Keep out of reach of children. *Jauhi daripada kanak-kanak.*

XVI. SHELF LIFE

Please refer to the expiry date on the outer carton.

XVII. PACKING / PACK SIZE(S)

Blister packs of 1 x 10's, 3 x 10's and 10 x 10's, in a box.

All pack sizes may not be available locally.

PRODUCT REGISTRATION HOLDER

Duopharma Marketing Sdn.Bhd.

Lot 2, 4, 6, 8 & 10

Jalan P/7, Section 13, Bangi Industrial Estate,

43650 Bandar Baru Bangi,

Selangor Darul Ehsan, Malaysia.

MANUFACTURER

Duopharma Manufacturing (Bangi) Sdn.Bhd.

Lot 2 & 4,

Jalan P/7, Section 13, Bangi Industrial Estate,

43650 Bandar Baru Bangi,

Selangor Darul Ehsan, Malaysia.